GL-ProQ-en-022

CTQ_CTP Management Procedure

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CTQ/CTP Management Procedure

ProQ_GL22

Purpose

The purpose of the CTQ/CTP management process is to ensure that all:

- Critical-to-Quality (CTQ) characteristics are collected and allocated to the process steps.
- Critical-to-Process (CTP) characteristics are prioritized and linked to the appropriate CTQ characteristics.
- The appropriate CTQs and CTPs are properly transferred to the Control Plan to keep in control Safety, Regulatory, and Performance-impacting characteristics.
- Variation of Critical-to-Quality (CTQ) and Critical-to-Process (CTP) characteristics is reduced before Release.

Scope

This procedure applies to all Schneider Electric (Hardware) **Offers** as well as all assemblies and internal group (IG) manufactured parts.

Target Audience

- Industrialization Quality Leader (IQL)
- Offer Industrialization Leader (OIL)
- Offer Technical Leader
- Industrialization Engineer
- Design Engineer Procurement
- Scrum Master
- Product Owner
- League Leader

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References

- **DirQ_GL41 Industrial Transformation Process Directive**
- **DirQ_GL43 Quality Fundamentals for GSC**
- **DirQ_GL44 Design for Safety & Reliability (DfSR) Directive**
- **ProQ_GL16 DfSR DFMEA Procedure**
- **ProQ_GL17 Process Failure Modes & Effects Analysis (PFMEA) Procedure**
- **ProQ_GL23 Derogation Management Procedure**
- **ProQ_GL30 Procedure DfSR Requirement & Allocation**
- **ProQ_GL32 DfSR Scorecard Procedure**
- **ProQ_GL33 Process Capability Study and Analyze Results Procedure**
- **ProQ_GL34 Control Plan Procedure**
- **ForQ_GL29 DfSR Scorecards Template**
- **ProQ_GL36 Pilot Run - PPAP Procedure**
- **ForQ_GL34 CTQ_CTP Management Workbook**
  - AIAG & VDA FMEA Handbook (August 2022)
  - **Internal Schneider Electric Encyclopedia iSEE**

Process Inputs

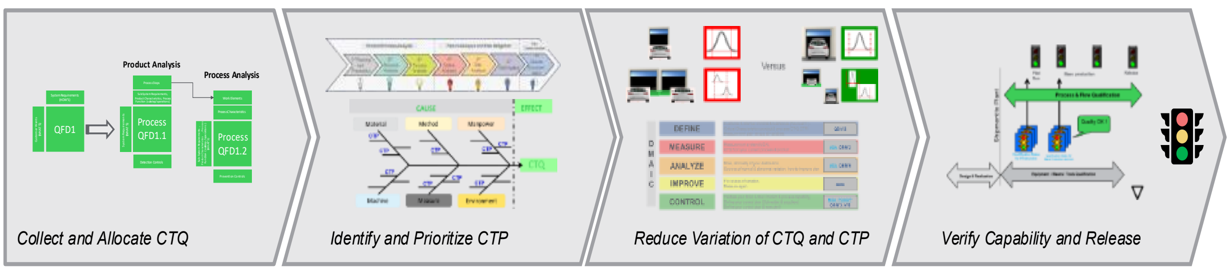
- Quality Functional Deployment of customer and stakeholder requirements to technical product requirements and product characteristics for each system level. (ref. **ProQ_GL30 Procedure DfSR Requirement & Allocation**)
- Product Characteristics their potential failure modes, effects, and Severities (ref. **ProQ_GL16 DfSR DFMEA Procedure**)
- Designation of CTQ Characteristics for each system level of assembly product in detailed design requirement documents such as mechanical drawings, functional assembly specifications, etc.
- Concept of Process Architecture or applicable changes

Process Overview

The CTQ/CTP Management process should initiate for each level of the system architecture as soon as possible and be concurrently developed with the product design of the respective system level to ensure timely manufacturability feedback and risk identification.

The CTQ/CTP Management Procedure (see process map in **Appendix A**) includes:

- **Collect and Allocate CTQ**
- **Identify and Prioritize CTP**
- **Reduce and Control Variation of CTQ and CTP**
- **Verify Capability and Release**



Collect and Allocate CTQ

The continuation of the House of Quality for the system, sub-system, and components to Process QFD enables the Industrialization team to maintain consistent prioritization of CTQ characteristics and evaluate the interactions of the respective system level to each Process Step.

It is important to collect all the relevant CTQs to ensure that the analysis of the interaction between the product needs and the designed process is comprehensive. Consider each level of the offer hierarchy as a potential source of CTQs.

The CTQ is then allocated to the appropriate Process Step(s). It is important to consider that:

- The point of creation is the first time the CTQ can be directly measured. If the point of creation is outside the process architecture (at a supplier) and not measured, the characteristic is “pass-through”.

- Process Steps before the point of creation of the CTQ may have product characteristics that contribute to the capability of the CTQ which can help in controlling the process. The team should consider what other documents should be updated to reflect these product characteristics (DFMEA, DfSR – QFD, Specifications, etc.)

- Process Steps after the point of creation of the CTQ may have the potential to change the CTQ.

As part of this analysis, document the current detection controls for the CTQ. Detection controls (in this context) are methods to measure, monitor, and react to data collected for the CTQ. (ref. **ProQ_GL17 Process Failure Modes & Effects Analysis (PFMEA) Procedure**)

Identify and Prioritize CTP

Once the CTQ characteristics have been allocated to the appropriate process steps, document the Process Work Elements for each process step that may interact with the product. For each work element, list the relevant process parameters and rate the strength of interactions to prioritize them and generate the CTP characteristics. As part of this analysis, document the current prevention controls for the CTQ (detection controls for CTPs).

It is recommended that the team utilize various "Generics" to support initial creation and ensure best practices are followed. (See **Appendix A**)

This step and the previous one, serve as a direct input to the Process Failure Modes and Effects Analysis (PFMEA). It supports Structure Analysis (Step 2), and Function Analysis (Step 3), as well as a portion of Risk Analysis (Step 5)

Reduce and Control Variation of CTQ and CTP

- Completion of the PFMEA Risk Analysis (Step 5) and Optimization (Step 6) require the use of problem-solving techniques such as DMAIC to develop and validate the manufacturing process proactively and robustly. (ref. **ProQ_GL17 Process Failure Modes & Effects Analysis (PFMEA) Procedure** and **ProQ_GL33 Process Capability Study and Analyze Results Procedure**)
    - Perform Measurement System Analyses to ensure the tools used to evaluate and control the assembly and process are robust and dependable for decision-making.
    - Perform Design of Experiments to establish the relationships between process characteristics and product characteristics.
    - Perform Design of Experiments to test response of CTQ's to setting of CTPs beyond the specified control limits.
- If, during the cross-functional development of the product and process, the process risks identified in the PFMEA are not mitigated to an acceptable level as defined by **ProQ_GL17 Process Failure Modes & Effects Analysis (PFMEA) Procedure**, the team <u>**shall**</u> transfer the CTQ's and CTPs to the production Control Plan to assure focus and attention to these characteristics.

Reduce and Control Variation of CTQ and CTP (cont.)

- Designation of CTQ or CTP S&R (Safety and Regulatory) **shall** be used in PFMEA, Control Plan and any Work Instructions if PFMEA Severity rating is 9 or greater **and** Occurrence rating is 4 or greater.

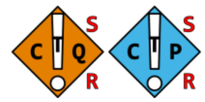


Figure 1

- Designation of CTQ or CTP (standard) shall be used in PFMEA, Control Plan and any Work Instructions if the PFMEA Severity rating is 7 **and** Occurrence rating is 7 or 10.

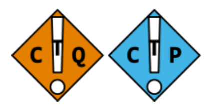


Figure 2

Verify Capability and Release

- Perform Measurement System Analysis and Capability Studies according to ProQ_GL33 Process Capability Study and Analyze Results Procedure to estimate the probability of exceeding specification limits (potential defect rate).

- Use SPC and Control Charting techniques on CTP's and CTQ's to provide early warning of potential instability, shift, or drift in the product or process. Use these tools to initiate problem-solving and avoid impacting customers and business performance.

- Follow ProQ_GL36 Pilot Run - PPAP Procedure to manage the release of the process.

- If the process is unable to comply with capability targets as identified in ForQ_GL29 DfSR Scorecards Template and ForQ_GL34 CTQ_CTP Management Workbook, the team shall consider:
    - Applying a TQW while process controls improvements are under development. (ref. DirQ_GL43 Quality Fundamentals for GSC)
    - Increasing the frequency of measurement, sample size, adding SPC controls, or 100% measurement

- Temporarily a derogation may be required. (ref. ProQ_GL23 Derogation Management Procedure.)

Process Outputs

- Capability data (raw data or mean and standard deviation) of CTQ characteristics for DfSR Scorecards
- Status of leading indicators in the CTQ/CTP Management workbook
- MSA, Capability, potential defect rate in the Process Scorecards

Monitoring & Control

The CTQ/CTP Management workbook (**ForQ_GL34**) enables tracking of leading indicators of the Process outputs (see **Appendix B**). The Industrialization Quality Leader **shall** monitor the following leading indicators for the applicable Offers:

- CTQs collected (not started, in-process, complete)
- CTQs allocated to process steps (not started, in-process, complete)
- # of CTQs or CTPs which are not: feasible, measurable, capable, or controlled

Roles & Responsibilities

Function	RACI Roles
Industrialization Quality Leader (IQL) or similar	R
Offer Industrialization Leader (OIL)	A
Design for Safety and Reliability Practitioner and/or Product Quality Leader	C
Offer Technical Leader	C
Industrialization Engineer	C
Design Engineer	C
Advanced Supplier Quality	C
Procurement	I
Scrum Master	I
Product Owner	I
League Leader	I

Competency Requirements

Role	Knowledge	Competency Level	Intervention
Process Owner	Process Subject Matter	Expert	Nomination from CS&Q QMS VP
Offer Industrialization Leader (OIL)	Application of Process on New Offer	Expert	Read Procedure, Training module, Workshop, Project, and Demonstrate Expert level competence
Industrialization Quality Leader (IQL) or similar	Application of Process	Advanced	Read procedure, Training module, Workshop, and Project
Advanced Supplier Quality	Application of Process	Advanced	Read procedure, Training module, Workshop, and Project
DfSR Practitioner and/or Product Quality Leader	Application of Process	Advanced	Read procedure, Training module, Workshop, and Project
Offer Technical Leader	Generation high quality inputs to the Process	Advanced	Read procedure, Training module, Workshop, and Project
Industrialization Engineer	Application of Process to Industrialization activities	Advanced	Read procedure, Training module, Workshop, and Project
Supply Chain Owner	Evaluation of the Supply Chain readiness (Master data, stock, supplier capacity).	Competent	Read procedure, Training module, Workshop
Procurement	Understanding of importance of the Process	Novice	Basic Overview Training Module and Read Procedure
Scrum Master	Understanding of importance of the Process	Novice	Basic Overview Training Module and Read Procedure
Product Owner	Understanding of importance of the Process	Novice	Basic Overview Training Module and Read Procedure
League Leader	Understanding of importance of the Process	Novice	Basic Overview Training Module and Read Procedure

Quality Records

Record	Record Location	Retention Period
CTQ/CTP Management Workbook	PCIM - Project Folder(s)	20 years

Terms & Acronyms

The table below includes definitions for terms and acronyms used in this document. Reference **iSEE** for a full list of terms and acronyms used in the SE IMS.

Acronyms	Definition
QFD	Quality Function Deployment
PFMEA	Process Failure Modes & Effects Analysis
DFMEA	Design Failure Modes & Effects Analysis
CTQ/CTP S/R	Critical to Quality/Critical to Process Safety & Regulatory
CTQ/CTP Standard	Critical to Quality/Critical to Process Standard
RPN	Risk Priority Number
TQW	Temporary Quality Wall
LOB	Line of Business
TRIZ	Theory of Inventive Problem Solving
SPC	Statistical Process Control

Terms	Definition
Characteristic	A distinguishing trait, quality, or property
Data	A collection of factual information (such as measurements or statistics) used as a basis for reasoning, discussion, or calculation
Attribute Data	Data which can be assigned to categorical or discrete variables
Variable Data	Data which can be assigned to a continuous or discrete variable (where the data has a predefined logical order)
Variable	A quantity that may assume any one of a set of values
Categorical Variable	Categorical variables contain a finite number of categories or distinct groups. Categorical data might not have a logical order. For example, categorical predictors include gender, material type, and payment method
Discrete Variable	Discrete variables are numeric variables that have a countable number of values between any two values. A discrete variable is always numeric. For example, the number of customer complaints or the number of flaws or defects

Terms	Definition
Continuous Variable	Continuous variables are numeric variables that have an infinite number of values between any two values. A continuous variable can be numeric or date/time. For example, the length of a part or the date and time a payment is received
Product Characteristic	A characteristic of a product is a distinctive feature or attribute that distinguishes it from other products in the same category
Process Characteristic	Process Characteristics are variables and parameters associated with elements of the process that have a cause- and- effect relationship with the variation found in product characteristics
Pass- through Characteristic	Pass Through characteristics are Product Characteristics that are not created within the scope of the process. Since they are not created within the process scope, they are not likely to be measured within the process scope, and therefore, is considered as "pass through" to the internal or external customers. Because of this, special considerations should be taken to ensure quality
Standard Characteristic	Other characteristics which are defined. If they are not maintained, the product or process will not likely be affected in a significant way. They are determined to be standard characteristics through requirements management, DFMEA, and PFMEA risk analysis and optimization steps (where severity and occurrence are defined)
Critical Characteristic	Customer Satisfaction depends on the capability of these characteristics. If they are not maintained, the product may not function up to our customer's expectations of our brand. They are determined to be critical characteristics through requirements management, DFMEA, and PFMEA risk analysis and optimization steps (where severity and occurrence are defined)
Regulatory Characteristics	Compliance with regulations depends on the capability of these characteristics. If they are not maintained, the product may not be legally sold. They are determined to be regulatory characteristics through requirements management, DFMEA, and PFMEA risk analysis and optimization steps (where severity and occurrence are defined)
Safety Characteristic	The safety of customers depends on the capability of these characteristics. If they are not maintained, the effects may be severe. They are determined to be safety characteristics through requirements management, DFMEA, and PFMEA risk analysis and optimization steps (where severity and occurrence are defined)
Critical to Quality	a subset of product characteristics that strongly relate to the customer's experience
Critical to Process	a subset of process characteristics that strongly relate to the CTQ or a subset of process characteristics that strongly relate to the process effectiveness

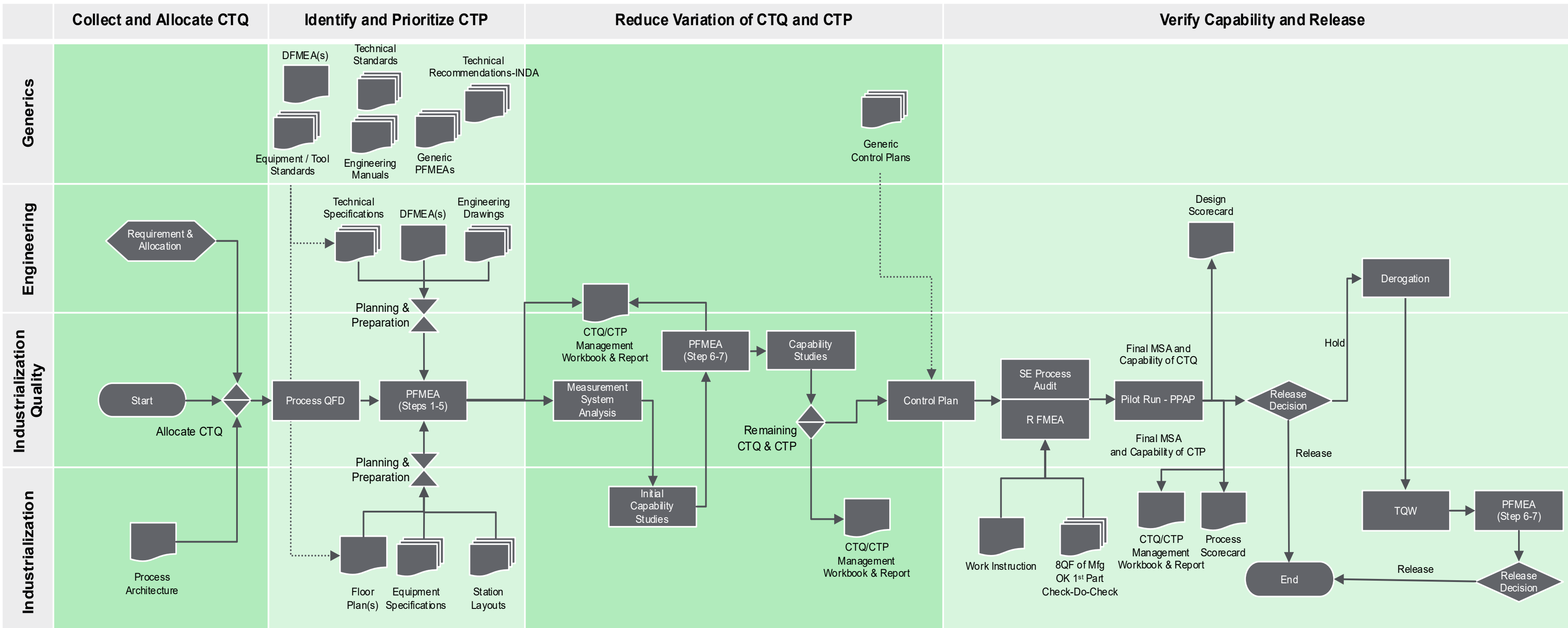
Terms	Definition
Structure Analysis	An analysis conducted to document the FMEA team's process hierarchy and set the basis/scope of the functional analysis includes the process flow and process hierarchy (structure tree)
Product (Structure analysis)	in the context of structure analysis, the Product represents the product, or its subassembly created by completing all the Process Steps within the hierarchy
Process Step (Structure analysis)	in the context of structure analysis, the Process Step is the specific manufacturing station where the process functions are applied to create the product, its subassemblies, or parts during the step to achieve the desired **product characteristics**
Quality Function Deployment	a focused methodology for carefully listening to the voice of the customer and then effectively responding to those needs and expectations

Document Control

Function	Names
Process Owner	Omar Ghani
Verifier/ Functional Leader	Eric Jacquot
Approver/ CS&Q QMS VP	Magdalena Michulec

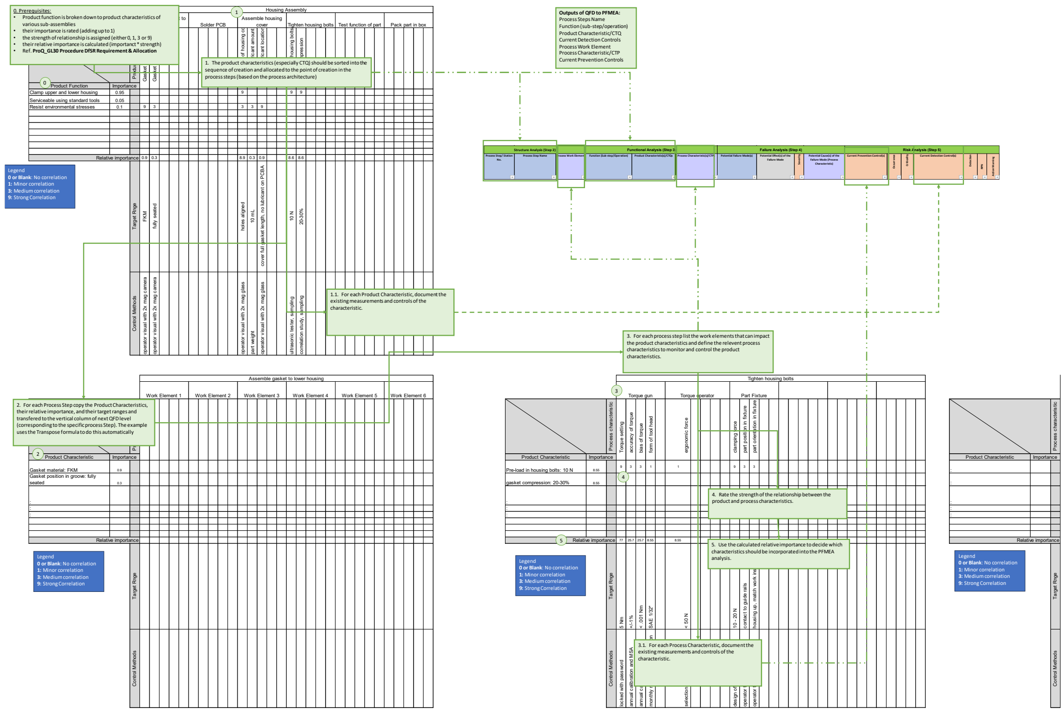
Revision	Date	Author	Summary of Change
1.0	Mar 21, 2024	O. Ghani	Initial release.

Appendix A: Process Map



Appendix B: CTQ/CTP Management Workbook

Process QFD



CTQ/CTP Management Worksheet & Report

CTQ/CTP Management Worksheet & Report

Offer name | Project / Epic Name | Release: Release ID by 3-Nov-23

Document Number:		Document Rev.:	
Offer Name:	Offer name	Create date:	3-Nov-23
Project / Epic Name:	Project / Epic Name	Last Update:	3-Nov-23
Planned Offer Release:	Release ID	Release date:	3-Nov-23

Quality Lead:	
Team:	

Risk / Feasibility Profile

						% Not Green	% Capable
Document Progress:	13%		R	0		33%	100%
			Y	1			
			G	2			
CTQ Reduced:	0	% Reduced	% Red. Doc'd	Pass-through:	6	% Pass-through	% Prevented
Controls to improve:	1	0%	100%	Prevention	2	67%	29%
				Detection	5		
Comments:							

ID	CTQ Level	CTQ / CTP	Characteristic Name	Risk / Feasi-bility	Gaps in document? DFMEA	Drawings	PFD	PFMEA	CP	JBS	Control Description frequency and chart type	Control Type	Reduction Rational Documented?	Pass-through?	Controls Improvement Needed?	Capable @ Pilot Run	Risk / Feasibility Explanation
1.0	Offer	CTQ S/R	Hole Diameter	G	Y						100% diameter verification - IMR chart	Detection		N	N	Y	
1.1		CTP	drill bit size		N							Detection		N	Y	Y	
1.2		CTP	machine feed		N						machine servo control	Prevention		N	N	Y	
1.3		CTP	machine speed		N						machine servo control	Prevention		Y	N	Y	
2.0	Part	CTQ	Material	G	N						supplier material certificate	Reduced	Y	Y	N	Y	
3.0	Sub-Assy	CTQ	weld strength	Y	Y	Y	Y	Y	Y	Y	5pcs sampling start and middle of shift - X-bar-R chart	Detection		Y	N	Y	
3.1		CTP	current		N							Detection		Y	N	Y	
3.2		CTP	time		N							Detection		Y	N	Y	
3.3		CTP	clamp force		N							Reduced	Y	Y	N	Y	

Scorecard

<<Enter Program Name Here>>	
Process Name	
Process Owner	
Today	7-Mar-24
Last Update	

User Inputs	Optional Inputs	Computed Values

Goal for Each CTQ/CTP within focused item shall be as follows:
- CTQ/CTPs with severity of 9/10: Cpk >=2.0,

Critical To Quality (CTQ/CTP) Factors	Severity	Type of data	Continuous Data: % GRR	Lower Spec Limit	Target Value	Upper Spec Limit	Sample Size	Mean Value	Standard Deviation	LCL	UCL	Cp	Cpk	Potential defect rate (%)	Discrete Data: % Appraiser vs. Standard	Number of Units	Number of Opportunity / Unit	Number of Defects	Total Opportunity	Defect per Opportunity	Defects per Million Opp.	Target: Target (Zst)	Target (Zlt)	Predicted: Predicted Z_{ST}	Predicted Z_{LT}
	10	Discrete													5%	9000	50	1	450000	0.000002	2	6.0	4.5	4.589	3.089
	9	Continuous	5%	5.5	10.0	13.5	30	10	1.5	5.50	14.50	2.00	0.78	111.65%								6.0	4.5	3.176	1.676
	7	Continuous	5%	5.5	10.0	13.5	30	10	10	-20.00	40.00	13.33	0.12	6895.25%								4.0	2.5	0.799	-0.701
	4	Continuous	5%	5.5	10.0	13.5	30	10	0.1	9.70	10.30	0.13	11.67	0.00%								3.0	1.5	11.500	10.000
	1	Continuous	5%	5.5	10.0	13.5	30	10	0.1	9.70	10.30	0.13	11.67	0.00%								3.0	1.5	11.500	10.000

REVISION HISTORY SUMMARY

Approvers	Requested By	Change Description	Reason for Change	Revision	Revision Date	Contributors
MAGDALENA MICHULEC; ERIC JACQUOT	OMAR GHANI	* This document has been migrated from the Global QMS Documents Platform (SharePoint) without revision changes. Process owner/Document author and revision details are specified in the document itself. * A summary of previous revisions is available only in the migrated version.	Document migration	1.0	03/21/2024	